WORKED SOLUTION

STEP III 2007, Question 7

Pure • Integration by Substitution

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s3-q7

The key idea

We are not allowed to say "arctan" or "π", so every fact has to come from changing variables inside the integrals. Each substitution is chosen so that the integrand $1/(1+u^2)$ (or $1/\sqrt{1-u^2}$) turns into itself or into the other one, with only the limits changing. Comparing limits then gives equations between values of t and s, and the symmetry $u \mapsto 1/u$ from part (i) is what turns those equations into multiples of p.

Solution

Part (i)

Take $x > 0$ and put $u = v^{-1}$, so $\mathrm{d}u = -v^{-2}\,\mathrm{d}v$. As u runs from 0 up to x, v runs from ∞ down to $1/x$. Also

$$\frac{1}{1+u^2} = \frac{1}{1+v^{-2}} = \frac{v^2}{v^2+1}.$$

Hence

$$t(x) = \int_{\infty}^{1/x} \frac{v^2}{1+v^2}\left(-\frac{1}{v^2}\right)\mathrm{d}v = \int_{1/x}^{\infty} \frac{1}{1+v^2}\,\mathrm{d}v,$$

where the minus sign was used to swap the limits. This is the required result. (The improper integral converges because the integrand is at most $1/v^2$.)

Applying this with $1/x$ in place of x gives $t(1/x) = \int_x^{\infty} \frac{1}{1+v^2}\,\mathrm{d}v$. Adding $t(x) = \int_0^x \frac{1}{1+v^2}\,\mathrm{d}v$, the two ranges join up to $[0,\infty)$:

$$t(x) + t(1/x) = \int_0^{\infty} \frac{1}{1+v^2}\,\mathrm{d}v = \tfrac{1}{2}p \qquad (x > 0).$$

Putting $x = 1$ gives $2t(1) = \frac{1}{2}p$.

Part (ii)

For $u \geq 0$ we have $0 \leq y < 1$. Squaring, $y^2(1+u^2) = u^2$, so $u^2(1-y^2) = y^2$ and

$$u = \frac{y}{\sqrt{1-y^2}},$$

taking the positive root because u and y have the same sign. By the quotient rule,

$$\frac{\mathrm{d}u}{\mathrm{d}y} = \frac{\sqrt{1-y^2} + \dfrac{y^2}{\sqrt{1-y^2}}}{1-y^2} = \frac{(1-y^2)+y^2}{(1-y^2)^{3/2}} = \frac{1}{\sqrt{(1-y^2)^3}},$$

as required.

Now substitute in $t(x)$. From $u^2 = y^2/(1-y^2)$ we get

$$1 + u^2 = \frac{1}{1-y^2},$$

so

$$\frac{1}{1+u^2}\,\frac{\mathrm{d}u}{\mathrm{d}y} = (1-y^2)\cdot\frac{1}{(1-y^2)^{3/2}} = \frac{1}{\sqrt{1-y^2}}.$$

The map $u \mapsto y$ is increasing (its inverse has positive derivative), with $u = 0 \mapsto y = 0$ and $u = x \mapsto y = x/\sqrt{1+x^2}$, which is less than 1. So

$$t(x) = \int_0^{x/\sqrt{1+x^2}} \frac{1}{\sqrt{1-y^2}}\,\mathrm{d}y = s\left(\frac{x}{\sqrt{1+x^2}}\right).$$

With $x = 1$ this reads $s(\frac{1}{\sqrt{2}}) = t(1) = \frac{1}{4}p$, using part (i).

Part (iii)

Write $a = \frac{1}{\sqrt{3}}$, so $z = \dfrac{u+a}{1-au}$. On $0 \leq u \leq a$ we have $au \leq \frac{1}{3}$, so the denominator stays positive and z is a smooth function of u there. Differentiating,

$$\frac{\mathrm{d}z}{\mathrm{d}u} = \frac{(1-au)+a(u+a)}{(1-au)^2} = \frac{1+a^2}{(1-au)^2}.$$

? **Ask yourself** Why should this substitution leave $1/(1+u^2)$ unchanged?

Compute $1+z^2$ over the common denominator. The cross terms $\mp 2au$ cancel:

$$1 + z^2 = \frac{(1-au)^2 + (u+a)^2}{(1-au)^2} = \frac{1 + a^2u^2 + u^2 + a^2}{(1-au)^2} = \frac{(1+a^2)(1+u^2)}{(1-au)^2}.$$

Dividing,

$$\frac{1}{1+z^2}\,\frac{\mathrm{d}z}{\mathrm{d}u} = \frac{1}{1+u^2},$$

so the integrand is unchanged and only the limits move. (Behind the scenes this is the tangent addition formula, but we don't need it.) When $u = 0$, $z = a = \frac{1}{\sqrt{3}}$. When $u = a$,

$$z = \frac{2/\sqrt{3}}{1 - \frac{1}{3}} = \frac{2}{\sqrt{3}} \cdot \frac{3}{2} = \sqrt{3}.$$

Therefore

$$t(\tfrac{1}{\sqrt{3}}) = \int_0^{1/\sqrt{3}} \frac{\mathrm{d}u}{1+u^2} = \int_{1/\sqrt{3}}^{\sqrt{3}} \frac{\mathrm{d}z}{1+z^2}.$$

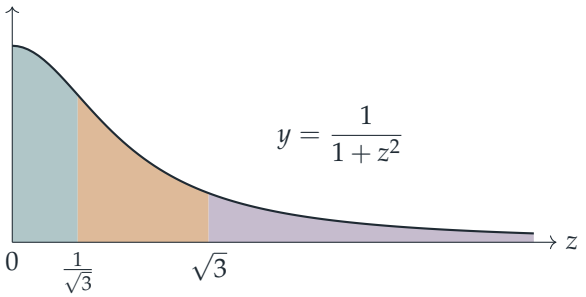
For the last step, split the range $[0, \infty)$ at $\frac{1}{\sqrt{3}}$ and $\sqrt{3}$:

$$\tfrac{1}{2}p = \int_0^{1/\sqrt{3}} \frac{\mathrm{d}z}{1+z^2} + \int_{1/\sqrt{3}}^{\sqrt{3}} \frac{\mathrm{d}z}{1+z^2} + \int_{\sqrt{3}}^{\infty} \frac{\mathrm{d}z}{1+z^2}.$$

The first piece is $t(\frac{1}{\sqrt{3}})$ by definition, and the second is $t(\frac{1}{\sqrt{3}})$ by what we have just shown. By part (i) with $x = \frac{1}{\sqrt{3}}$, the third piece is $\int_{1/x}^{\infty} \frac{\mathrm{d}z}{1+z^2} = t(x) = t(\frac{1}{\sqrt{3}})$ as well. Hence

$$3\,t(\tfrac{1}{\sqrt{3}}) = \tfrac{1}{2}p.$$

The picture shows the three pieces: the area under $1/(1+z^2)$ from 0 to ∞ splits into three regions of equal area.



Checks

Numerically (Simpson's rule), $t(\frac{1}{\sqrt{3}}) \approx 0.5235988$ and $\int_{1/\sqrt{3}}^{\sqrt{3}} \frac{\mathrm{d}z}{1+z^2} \approx 0.5235988$, while $p \approx 3.1415927$, so $3t(\frac{1}{\sqrt{3}}) \approx 1.5707963 \approx \frac{1}{2}p$. Likewise $s(\frac{1}{\sqrt{2}}) \approx t(1) \approx 0.7853982 \approx \frac{1}{4}p$. Of course p is really π, and these say $\arctan\frac{1}{\sqrt{3}} = \frac{\pi}{6}$, $\arcsin\frac{1}{\sqrt{2}} = \arctan 1 = \frac{\pi}{4}$; you may use that to check, though not in the answer.

Common slips

- Forgetting that $u = 1/v$ reverses the direction of integration; the minus sign in $\mathrm{d}u$ is what puts the limits back in order.

- In part (iii), using the substitution and then stopping: the integral from $\frac{1}{\sqrt{3}}$ to $\sqrt{3}$ is only one of three equal pieces, and part (i) is needed for the piece beyond $\sqrt{3}$.
- Not checking that $1 - au \neq 0$ on the range; here $au \leq \frac{1}{3}$, so the substitution is valid.

Beyond STEP

This question is a glimpse of how analysts define the trigonometric functions rigorously: one defines $\arctan x$ as $\int_0^x \frac{\mathrm{d}u}{1+u^2}$, defines π as twice its limit at infinity, and derives every property from substitutions like these, without any geometry. The substitution in part (iii) is the addition formula for tan in disguise, and the same method proves $t(x) + t(y) = t\left(\frac{x+y}{1-xy}\right)$ for $xy < 1$. Abbott's *Understanding Analysis* shows how integrals and power series are used to put such definitions on a firm footing.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.